Synthetic RGB-D Data Generation for Robotic Perception in Manufacturing Environments

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***Abstract*—Synthetic RGB-D data can support the development of robotic perception systems, but ideal rendered depth does not reproduce the measurement imperfections of physical cameras. This work presents a procedural pipeline for generating labelled synthetic RGB-D data of a manufacturing scene containing aluminium profiles and mounting brackets. The Intel RealSense D455 is selected as a reference camera through a qualitative multi-criteria assessment. A phenomenological stereo-inspired depth model introduces disparity noise and quantisation, as well as range-dependent and depth-discontinuity-related missing measurements. A randomized dataset of 200 object poses is generated with corresponding pixel-level labels and pose metadata.**

Two downstream geometric estimators are investigated. A PCA-based oriented bounding-box estimator exhibits a median translation error increase from 8.42 mm with ideal depth to 11.55 mm under the complete corruption model. In a paired ablation, disparity noise accounts for most of this degradation, whereas the effect of the implemented dropout mechanisms is comparatively small. A spatial-correlation sensitivity analysis found that increasing the correlation scale to 16 pixels, relative to independent noise, reduced mean cross-axis error by 1.07 mm while increasing mean axis error by 0.050 degrees; these effects were statistically detectable but small and estimator-specific. Under a strict 5 mm total-translation and 5-degree axis tolerance, its geometric success rate decreases from 18.0% to 0.4%. A complementary CAD-informed end-face estimator returns pose estimates in approximately 68–69% of trials, with conditional median translation errors between 2.23 and 2.46 mm. Its principal failure mode is insufficient visible end-face evidence, which cannot be resolved simply by relaxing acceptance thresholds. The sensor model is not calibrated against a physical D455, and both estimators use ground-truth segmentation and camera parameters. The results therefore provide a controlled synthetic investigation of sensor-model sensitivity and geometric perception, rather than evidence of physical grasp performance or general sim-to-real transfer.

***Index Terms*—RGB-D sensing, synthetic data, depth simulation, robotic perception, 3D vision, manufacturing.**

I. Introduction

Robotic manipulation in manufacturing environments often requires accurate perception of objects with complex geometry, reflective surfaces, and partially occluded features. Aluminium extrusion profiles and mounting brackets are representative examples: their geometric structure is suitable for automated assembly, but reliable localisation and grasping depend on the quality of the available visual and depth measurements.

Synthetic data offers a practical way to generate large amounts of labelled training and evaluation data without manually annotating every observation. Procedural rendering also provides access to ground-truth geometry, camera parameters, object poses, and semantic labels. BlenderProc provides a programmable framework for generating such annotated scenes [1].

However, depth maps produced by conventional rendering pipelines typically represent ideal geometry and do not automatically reproduce the measurement imperfections of a physical RGB-D camera. A perception system developed using only ideal synthetic depth may therefore encounter a domain gap when deployed on a real robot. Triangulation-based depth sensing, including stereo, depends on image correspondence and can exhibit range-dependent uncertainty, quantisation, missing measurements, and failures around occlusions and depth discontinuities [2], [3].

This assignment develops a procedural manufacturing scene and randomized RGB-D dataset, using the Intel RealSense D455 as a reference sensor. A phenomenological depth model approximates selected stereo measurement imperfections, and two geometric pose estimators are used to investigate their downstream effects. The study also identifies key sensor-model limitations and priorities for real-camera validation.

II. RGB-D Camera Selection

Five commercially available RGB-D cameras were considered: the Intel RealSense D455 [4], Orbbec Gemini 2 [5], Orbbec Femto Bolt [6], Photoneo MotionCam-3D M [7], and Zivid 2+ MR60 [8]. These candidates represent different sensing technologies and trade-offs in measurement range, spatial resolution, software accessibility, and depth characteristics.

The candidates were evaluated using seven criteria: depth-error characterisation (20%), feasibility of phenomenological modelling (25%), SDK and data accessibility (20%), RGB–depth integration (10%), manufacturing relevance (10%), spatial resolution and range (5%), and documentation and reproducibility (10%). Each criterion was assigned a score from 1 (poor) to 5 (excellent), based on a qualitative assessment of publicly available specifications and documentation. These scores represent suitability for this simulation task, not experimentally measured camera performance.

Table I presents the resulting weighted scores. The criteria are abbreviated as follows: DE: depth-error characterisation; PM: phenomenological modelling; SDK: software accessibility; RGB: RGB–depth integration; MFG: manufacturing relevance; SR: spatial resolution and range; and REP: reproducibility.

TABLE I
QUALITATIVE WEIGHTED CAMERA-SELECTION SCORES.

Camera	DE	PM	SDK	RGB	MFG	SR	REP	Total
D455	4	5	5	4	3	3	5	**4.40**
Gemini 2	3	4	4	4	3	3	4	3.65
Femto Bolt	4	4	4	4	3	3	5	3.95
Zivid 2+	5	3	4	5	3	3	5	4.00
Photoneo	4	3	3	5	3	3	4	3.50

The total is the weighted sum, normalised to a maximum of 5.

The Intel RealSense D455 achieved the highest weighted score (4.40/5). Its stereo-based depth sensing, accessible RGB, depth and infrared streams, calibration information, and open-source RealSense SDK make it a practical reference for developing a reproducible phenomenological model. The D455 has a published operating range of approximately 0.6–6 m, a depth field of view of approximately 86–87 degrees horizontally and 57–58 degrees vertically, and a stereo baseline of 95 mm [4], [9]. The rendering configuration uses a lower resolution of 640×360 pixels.

The weighting deliberately prioritises modelling feasibility and software accessibility, which together account for 45% of the total score. Manufacturing relevance contributes 10%, and all candidates received the same qualitative score for this criterion. The ranking is therefore specific to the objective of building an accessible and reproducible sensor simulation, rather than an objective ranking of industrial measurement performance.

The ranking is task-dependent. Increasing depth-error characterisation to 30% and manufacturing relevance to 20%, while reducing modelling feasibility and SDK accessibility to 10% each, and setting RGB–depth integration, spatial resolution and range, and reproducibility to 10% each, makes the Zivid 2+ the preferred candidate (4.10 versus 4.00 for the D455). The D455 is therefore selected for reproducibility and modelling accessibility, not industrial metrology. Structured-light systems may be more suitable for reflective metal surfaces in production.

The D455 selection does not imply that its measurement characteristics have been reproduced or calibrated in this work. No physical D455 was available. Its public specifications inform the camera geometry and the choice of a stereo-inspired model, while the corruption parameters remain explicit simulation assumptions.

III. SYNTHETIC SCENE AND DATASET GENERATION

A. Procedural Manufacturing Scene

A procedural manufacturing scene was developed using BlenderProc [1]. It contains aluminium extrusion profiles, mounting brackets, and a workbench. The extrusion profiles include simplified surface grooves to approximate the appearance of industrial T-slot profiles. These grooves are represented using additional geometry rather than true Boolean cuts.

The scene geometry, camera configuration, and lighting are defined programmatically. Semantic labels are assigned to the workbench, aluminium profile geometry, mounting brackets, and the designated target extrusion. The category IDs are 0 for background or unlabelled pixels, 1 for the workbench, 2 for other profile geometry, 3 for mounting brackets, and 4 for the target extrusion.

The rendering resolution is 640×360 pixels. The camera intrinsics are derived from the approximate D455 field of view:

$$\mathbf{K} = \begin{bmatrix} 343.158 & 0 & 319.5 \\ 0 & 331.519 & 179.5 \\ 0 & 0 & 1 \end{bmatrix}. \tag{1}$$

Since the published horizontal and vertical fields of view are approximate, deriving the two focal lengths independently also introduces an approximately 3.5% difference between the horizontal and vertical focal lengths in pixel units. The resulting intrinsics are retained consistently throughout the synthetic experiments, but should not be interpreted as a calibrated D455 model.

These are rendering intrinsics, not measured physical D455 calibration parameters. The intrinsic matrix, camera pose, target dimensions, and object transforms are saved in ground-truth metadata alongside the rendered data.

The target extrusion is represented by a simplified cuboid with half-extents of $[0.65, 0.04, 0.04]$ m. This model provides a controlled geometric reference, but it is not a complete CAD representation of a commercial T-slot extrusion.

Figure 1 shows the rendered RGB image, ideal depth map, and semantic segmentation.

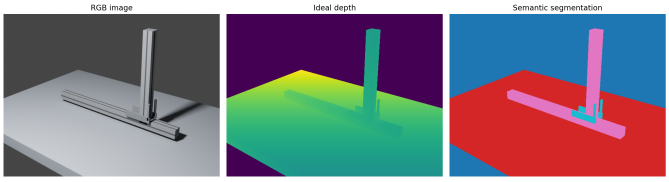


Fig. 1. Procedurally generated manufacturing scene showing RGB, ideal rendered depth, and semantic segmentation.

B. Randomized Pose Dataset

To evaluate geometric perception beyond a single fixed object configuration, a dataset of 200 randomized target poses was generated. The extrusion yaw was sampled uniformly from $[-\pi, \pi]$, and its horizontal translations were sampled independently from $[-0.06, 0.06]$ m along each of the world x and y axes. The target height was fixed at $z = 0.12$ m. The camera pose, camera intrinsics, scene geometry, and lighting configuration were held fixed.

The generator uses random seed 20261104. Each sample contains an RGB image, ideal depth map, semantic segmentation, and metadata including the target-to-world transformation, assembly transformation, camera pose, and target pixel and valid-depth counts. The 200 rendered samples are stored as individual HDF5 files rather than as a single image stack. A small subset of three samples is included in the repository

as a reproducible example; the full dataset can be regenerated using the provided script.

All 200 samples contain target pixels and valid target depth measurements. The number of target pixels ranges from 1,145 to 4,950, with a median of 3,983.5. The variation reflects changes in visible projected geometry and partial occlusion under the randomized poses.

Label consistency was checked by back-projecting target pixels and transforming them into the labelled target frame. No reconstructed points were more than 5 mm outside the target bounding geometry. The median per-sample median surface deviation was 0.0018 mm, and the maximum per-sample 95th-percentile deviation was 0.0034 mm. This verifies internal consistency between the renderer, coordinate transformations, and saved labels, not physical geometric or sensor accuracy.

C. Stereo Viewpoint Visibility Audit

A reprojection audit using virtual camera positions separated by the assumed 95 mm baseline found that most target points were geometrically visible from both viewpoints. This does not establish successful stereo correspondence, which also depends on texture, reflectance, occlusion, and matching behaviour. The audit therefore provides a geometric diagnostic, not a physical estimate of stereo depth completeness.

IV. Phenomenological Depth Sensor Model

A. Stereo-Inspired Disparity Noise

The ideal depth maps generated by BlenderProc represent rendered geometry rather than noisy stereo measurements. A separate processing stage therefore introduces selected measurement imperfections inspired by the D455 stereo architecture.

For a rectified stereo pair, depth is related to disparity by

$$z = \frac{fB}{d}, \tag{2}$$

where f is focal length in pixels, B is the stereo baseline, and d is disparity in pixels. The model uses $f = 343.158$ pixels and $B = 0.095$ m.

Instead of adding noise directly in depth, the model perturbs disparity:

$$d' = d + \epsilon_d, \qquad \epsilon_d \sim \mathcal{N}(0, \sigma_d^2), \tag{3}$$

followed by disparity quantisation:

$$d_q = q \cdot \mathrm{round}\left(\frac{d'}{q}\right), \qquad q = \frac{1}{32} \text{ pixel}. \tag{4}$$

The corrupted depth is recovered as

$$z' = \frac{fB}{d_q}. \tag{5}$$

The nominal disparity-noise standard deviation is $\sigma_d = 0.06$ pixels. Non-positive or otherwise invalid reconstructed measurements are marked as missing.

This formulation captures the nonlinear relationship between disparity uncertainty and depth uncertainty: a similar disparity perturbation produces a larger depth error at greater distances. The parameter values are modelling assumptions, not measurements from a physical D455. The model does not reproduce the D455's complete stereo matching or post-processing pipeline.

B. Range-Dependent Missing Measurements

A range-dependent probability introduces additional invalid measurements:

$$p_{\mathrm{range}}(z) = 0.002 + 0.03\left(\frac{z}{6}\right)^2. \tag{6}$$

This represents a baseline probability of missing measurements with an additional distance-dependent component. It is an assumed phenomenological function rather than a fitted camera characteristic.

C. Depth-Discontinuity-Related Failures

Stereo matching can become less reliable near occlusions and depth discontinuities. To approximate a subset of these effects, the segmentation map is examined for adjacent pixels with different semantic labels. A boundary failure is considered only where the corresponding depth difference is at least 50 mm. No morphological dilation is applied. A probability of 0.05 is used for pixels selected by this boundary mask.

The range and boundary failure events are sampled independently. Their combined probability is therefore

$$p_{\mathrm{invalid}}(z, e) = 1 - (1 - p_{\mathrm{range}}(z))(1 - 0.05e), \tag{7}$$

where $e = 1$ denotes a pixel selected by the depth-discontinuity mask and $e = 0$ otherwise. Invalid measurements are represented by NaN values. Independent random-number streams are used for the corruption components to support reproducible ablations.

The boundary model is deliberately conservative in scope. It is not a physical stereo-occlusion simulator: semantic transitions and depth jumps are only proxies for locations where matching may be difficult. In particular, a semantic boundary does not necessarily imply stereo occlusion, and stereo failures can also occur within a single semantic region.

D. Depth Corruption Examples

Figure 2 illustrates the effect of the proposed corruption model on a rendered depth image. The visualisation separates valid noisy measurements from pixels invalidated by the simulated missing-measurement mechanisms.

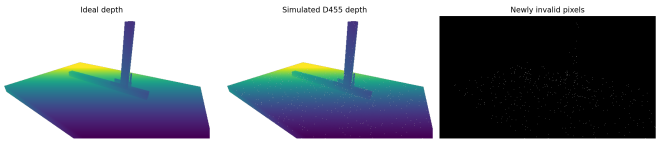


Fig. 2. Illustrative D455-inspired depth corruption. Disparity-domain noise, range-dependent missing measurements, and depth-discontinuity-related failures are applied to the rendered depth image. Invalid pixels are shown separately from valid noisy measurements. The model is phenomenological and is not fitted to physical D455 measurements.

E. Single-Scene Sensor Evaluation and Ablation

The sensor model was evaluated on the fixed manufacturing scene. Of 101,523 originally valid depth pixels, 101,104 remained valid after full corruption, corresponding to an invalidation rate of 0.41%. On commonly valid pixels, the depth MAE was 3.01 mm and RMSE was 4.36 mm. Range-stratified RMSE increased from 3.66 mm at 0–2 m to 9.73 mm at 2–3 m, consistent with the nonlinear disparity-to-depth relationship.

TABLE II
DEPTH ERROR AFTER DISPARITY CORRUPTION, EVALUATED AGAINST THE CORRESPONDING RENDERED DEPTH AT DIFFERENT RANGES.

Depth range	Valid points	Invalid (%)	MAE (mm)	RMSE (mm)
0–2 m	94,430	0.39	2.67	3.66
2–3 m	7,093	0.69	7.58	9.73

As expected from the inverse-disparity depth relationship, the same disparity perturbation produces larger metric depth errors at greater distances. The reported values characterise the implemented phenomenological model, rather than measured D455 performance.

An ablation separated disparity perturbation from the two missing-measurement mechanisms. Disparity-only corruption produced an RMSE of 4.37 mm, while range- and boundary-only dropout invalidated 0.36% and 0.06% of pixels, respectively, without changing the values of retained measurements. The full model invalidated 0.41% of pixels.

TABLE III
ABLATION OF THE SIMULATED DEPTH-CORRUPTION COMPONENTS. ERROR METRICS ARE COMPUTED OVER RETAINED VALID MEASUREMENTS.

Condition	Invalid (%)	MAE (mm)	RMSE (mm)	Bias (mm)
Disparity noise only	0.00	3.02	4.37	0.02
Range missing only	0.36	0.00	0.00	0.00
Boundary failures only	0.06	0.00	0.00	0.00
Full corruption	0.41	3.01	4.36	0.02

The zero error values for the dropout-only conditions do not indicate perfect depth estimation: these conditions remove measurements without perturbing the depths that remain valid. The full-corruption invalid fraction is not exactly the sum of the individual dropout fractions because the two dropout mechanisms can affect overlapping pixels. These results describe the selected synthetic scene and assumed parameters, not measured D455 performance.

F. Sensitivity to Spatial Correlation

The primary sensor model uses independent Gaussian disparity perturbations. However, real stereo matching errors may exhibit spatial structure. To investigate whether the downstream results depend on this modelling choice, an additional sensitivity experiment compared independent noise with spatially correlated disparity fields.

The correlated fields were generated by smoothing Gaussian white noise with a Gaussian filter at four spatial scales: 2, 4, 8, and 16 pixels. For each correlated field, the values over valid depth pixels were normalised to zero mean and a standard deviation of 0.06 pixels before disparity quantisation. The independent condition used per-pixel Gaussian samples with the same nominal standard deviation. All conditions used a disparity quantisation interval of 1/32 pixel. Range-dependent and depth-discontinuity-related dropout were disabled to isolate the effect of spatial correlation. The experiment used the same 200 randomized poses and five repetitions per pose for each condition, giving 5,000 trials in total. The correlation scales are phenomenological sensitivity parameters; they were not measured from a physical D455.

TABLE IV
SENSITIVITY OF THE PCA-BASED ESTIMATOR TO DISPARITY-NOISE CORRELATION SCALE.

Condition	Med. $e_\perp$ (mm)	P95 $e_\perp$ (mm)	Med. e_a (°)	P95 e_a (°)	Success (%)
Independent	10.885	21.148	0.430	1.579	1.2
Corr., 2 px	10.519	21.066	0.433	1.571	3.0
Corr., 4 px	10.504	21.127	0.447	1.581	4.6
Corr., 8 px	10.345	21.111	0.476	1.588	7.1
Corr., 16 px	10.027	21.206	0.505	1.583	11.0

The pre-quantisation disparity standard deviation was approximately 0.06 pixels across all conditions. Table IV shows that increasing the correlation scale reduced median cross-axis error from 10.88 to 10.03 mm, while increasing median axis error from 0.430 to 0.505 degrees. Geometric success increased from 1.2% to 11.0% under the selected 5 mm cross-axis and 5-degree orientation tolerances. All conditions achieved 100% estimator coverage.

Unlike the earlier OBB ablation, this sensitivity analysis uses cross-axis translation error rather than total translation error for its geometric success criterion, reflecting the distinction between positioning along the extrusion and positioning perpendicular to its longitudinal axis.

Paired pose-level bootstrap comparisons confirmed these trends: at a 16-pixel scale, mean cross-axis error decreased by 1.07 mm (95% CI: $[-1.26, -0.89]$ mm), while mean axis error increased by 0.050 degrees (95% CI: [0.039, 0.060]). These statistically detectable changes are small in absolute magnitude and are specific to this estimator; they do not indicate that correlated noise improves physical camera performance.

These results demonstrate an estimator-specific sensitivity to spatial correlation. Since none of the tested scales has been calibrated against D455 measurements, the independent-noise model remains the primary sensor model; correlated cases are included only as a sensitivity analysis.

Figure 3 visualises the paired changes in cross-axis and orientation error relative to independent noise.

G. Target-Specific Dropout and Stereo Visibility

Of 3,879 target pixels, 441 (11.37%) fell within the depth-discontinuity mask. Across ten random seeds, the range-only, boundary-only, and combined mechanisms invalidated

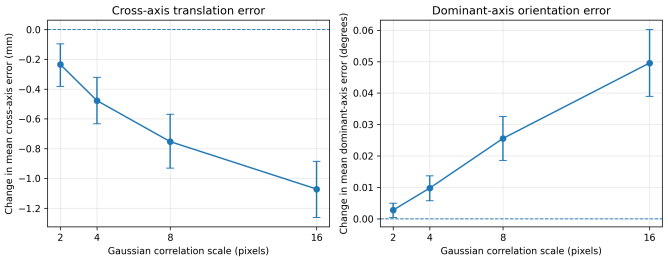


Fig. 3. Effect of spatially correlated disparity noise on OBB pose estimation, relative to the independent-noise reference. Left: change in mean cross-axis translation error. Right: change in mean dominant-axis orientation error. Error bars indicate 95% paired pose-level bootstrap confidence intervals over 200 poses.

approximately 0.40%, 0.61%, and 1.00% of target pixels, respectively. A geometric reprojection audit using virtual views at ± 95 mm found 99.92% and 99.56% of target pixels visible and consistent in the two views. As this audit does not simulate stereo correspondence, it does not establish physical stereo visibility; however, it provided no justification for adding a separate stereo-occlusion dropout term based on the semantic-edge heuristic.

V. Downstream Geometric Perception

A. Point-Cloud Reconstruction

Pixels belonging to the target extrusion (category ID 4) are selected using the ground-truth semantic segmentation. For a pixel (u, v) with depth z, the point is back-projected into the computer-vision camera coordinate frame:

$$\begin{aligned} X &= \frac{(u - c_x)z}{f_x}, \\ Y &= \frac{(v - c_y)z}{f_y}, \\ Z &= z. \end{aligned} \tag{8}$$

The resulting points are transformed into the world frame using the saved camera pose. The same segmentation and camera parameters are used for the ideal and corrupted conditions, so the depth map is the only perception input changed in the paired comparison.

This setup isolates depth corruption, but also provides privileged information that would not generally be available to a deployed perception system. In particular, the estimator does not need to detect or segment the object, estimate camera intrinsics, or recover the camera pose.

B. PCA-Based Oriented Bounding-Box Estimator

A PCA-based oriented bounding-box (OBB) estimator was implemented as an interpretable geometric baseline. Given the reconstructed target point cloud, the points are centred and a principal-component analysis is performed. The dominant eigenvector provides an estimate of the extrusion's longitudinal axis. The points are projected into the principal-component coordinate frame, and the midpoint of their axis-aligned bounding box is transformed back into the world frame to estimate the object centre.

The estimator therefore returns a translation estimate and a sign-invariant estimate of the dominant object axis. It does not recover a complete six-degree-of-freedom orientation. In particular, the two cross-sectional axes of the simplified extrusion are not reliably determined by this PCA construction, and the sign of the longitudinal axis is treated as ambiguous.

Translation error is defined as

$$e_t = \|\mathbf{t}_{\text{est}} - \mathbf{t}_{\text{gt}}\|_2 , \tag{9}$$

and the sign-invariant dominant-axis error is

$$e_a = \cos^{-1}\left(\left|\mathbf{a}_{\text{est}}^{\top}\mathbf{a}_{\text{gt}}\right|\right), \tag{10}$$

where both axis vectors are unit-normalised. A trial is classified as a successful geometric grasp-pose proxy if translation error is at most 5 mm and dominant-axis error is at most 5 degrees. This is an illustrative tolerance, not a physically validated grasp-success criterion.

C. Randomized Pose Experiment

The estimator was evaluated on the 200 randomized target poses described in Section III. Each pose was evaluated once with ideal depth and five times with independently corrupted depth. This resulted in 200 ideal trials and 1,000 full-corruption trials.

The ideal-depth baseline achieved a median translation error of 8.42 mm (P95: 23.05 mm). Under full corruption, these values increased to 11.55 mm and 25.50 mm, respectively. Median dominant-axis error remained approximately 0.43 degrees, while the mean invalid target-depth rate was 0.88%. The strict 5 mm translation and 5-degree axis criterion was satisfied in 18.0% of ideal trials and 0.4% of corrupted trials. The low ideal-depth success rate indicates that visibility and estimator limitations already contribute substantially before sensor corruption.

D. Paired Corruption Ablation

To identify which corruption components contribute most to the downstream change, a paired ablation was performed over the same 200 randomized poses. Ideal depth was evaluated once per pose, while each corrupted condition was evaluated using five independent random seeds per pose. The three corruption conditions were disparity noise only, dropout only, and full corruption, giving 3,200 trials in total.

Table V reports median total translation (e_t), cross-axis ($e_\perp$) and along-axis ($e_\parallel$) errors, dominant-axis error (e_a), and geometric success rate. Success uses a 5 mm cross-axis and 5-degree axis tolerance.

A decomposition of translation error showed that cross-axis error dominates the ideal-depth error, while along-axis error also increases under full corruption, from 0.35 mm to 2.13 mm. The persistent cross-axis error is consistent with bias in the estimated box centre when only camera-facing surfaces are observed. Under full corruption, median cross-axis error

TABLE V
RANDOMIZED-POSE OBB RESULTS. SUCCESS IS BASED ON A 5 MM CROSS-AXIS AND 5-DEGREE AXIS TOLERANCE. CORRUPTED CONDITIONS USE FIVE TRIALS PER POSE.

Condition	e_t (mm)	$e_\perp$ (mm)	$e_\parallel$ (mm)	e_a (°)	Success (%)
Ideal	8.42	8.30	0.35	0.432	18.0
Disparity	11.40	10.78	2.19	0.430	1.0
Dropout	8.57	8.42	0.37	0.427	17.3
Full	11.55	10.94	2.13	0.426	1.0

increased from 8.30 mm to 10.94 mm. Using a 5 mm cross-axis tolerance and a 5-degree axis tolerance, geometric success decreased from 18.0% to 1.0%, compared with 18.0% to 0.4% under the total-translation criterion. The two criteria represent different positioning requirements.

The paired analysis calculates a mean translation-error change for each pose before aggregating across the 200 poses. Disparity noise increased mean translation error by 2.23 mm relative to ideal depth, with a bootstrap 95% confidence interval of [1.97, 2.50] mm. Dropout-only corruption increased it by 0.04 mm [0.004, 0.084] mm. The complete model increased it by 2.28 mm [2.01, 2.55] mm. The bootstrap resamples poses, rather than individual repeated trials, to preserve the pose as the independent experimental unit.

The corresponding mean change in dominant-axis error was close to zero for all three corruption conditions. Under the total-translation criterion, the success rate decreased by 17.6 percentage points under disparity-only and full corruption, compared with a 0.8 percentage-point decrease under dropout-only corruption. These results identify disparity perturbation as the dominant source of degradation for this estimator and this scene. They do not establish that missing measurements are unimportant in general: the result depends on the particular point-cloud estimator, target geometry, and missing-data pattern.

E. Sensitivity to the Grasp Tolerance

The strict 5 mm translation threshold makes the binary success metric sensitive to the chosen tolerance. To investigate this dependence, the translation threshold was varied from 5 to 25 mm, while the dominant-axis threshold was kept fixed at 5 degrees.

Figure 4 shows the resulting geometric grasp-feasibility rates across the four experimental conditions. At a 10 mm translation tolerance, success was 55.5% for ideal depth, 38.5% for disparity-only corruption, 55.4% for dropout-only corruption, and 38.2% for full corruption. At 20 mm, the respective success rates were 94.0%, 85.6%, 94.0%, and 85.6%.

Disparity noise is the dominant source of degradation, whereas dropout-only results remain close to the ideal baseline. The results also show that the measured degradation depends on the assumed application tolerance. These thresholds are illustrative and have not been validated using a physical gripper or assembly task. Note that this tolerance sweep uses total translation error, whereas the geometric success column in Table V uses cross-axis translation error. The two criteria should not be compared directly.

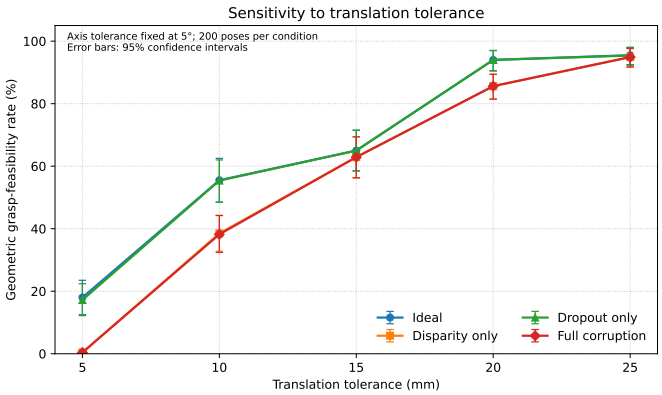


Fig. 4. Sensitivity of geometric grasp-feasibility to the total-translation tolerance. The dominant-axis tolerance is fixed at 5 degrees. Error bars indicate 95% confidence intervals across 200 randomized poses per condition.

F. CAD-Informed End-Face Pose Estimation

The limitations of the PCA-OBB estimator motivated a second, more task-informed approach. Rather than estimating the object centre from the midpoint of the visible point-cloud bounding box, this method uses the known geometry of the simplified target and attempts to identify one of its end faces.

The estimator first determines the longitudinal object axis using robust principal-component analysis. Surface normals are then estimated from the reconstructed point cloud. Points whose horizontal normals are sufficiently aligned with the estimated longitudinal axis are selected as potential end-face observations. These points are projected onto the axis and separated into spatial clusters. Candidate clusters are fitted with planes, and the estimated object centre is recovered by offsetting the selected end-face position by the known half-length of the target (0.65 m).

The experiment used the same 200 randomized poses, with one ideal-depth trial and three repetitions per corrupted condition (2,000 trials total). The estimator returned a pose in approximately 68–69% of trials. Table VI summarises coverage and pose errors. Error statistics are conditional on a returned estimate; coverage counts all trials.

TABLE VI
CAD-INFORMED END-FACE ESTIMATOR RESULTS.

Condition	Cov. (%)	Med. e_t (mm)	P95 e_t (mm)	Med. e_a (°)	P95 e_a (°)
Ideal	68.5	2.46	7.27	1.205	2.106
Disparity	68.8	2.27	7.55	1.156	2.076
Dropout	68.0	2.23	6.74	1.188	2.133
Full	68.7	2.24	7.22	1.146	2.083

The slightly lower conditional median errors in some corrupted conditions should not be interpreted as an improvement caused by sensor noise: the returned-estimate populations

differ because the estimator's acceptance and failure patterns change.

Under the illustrative 5 mm translation and 5-degree axis tolerance, the success rates were 60.0% for ideal depth, 58.3% for disparity-only corruption, 61.0% for dropout-only corruption, and 58.2% for full corruption. At a 10 mm translation tolerance, the corresponding rates were 68.5%, 67.0%, 67.7%, and 67.0%. These results show relatively small differences between the sensor conditions for this estimator. They do not demonstrate that the method is insensitive to realistic sensor errors, since the simulated corruption model is limited and the estimator uses privileged geometric information.

1) Diagnosis of End-Face Estimation Failures: Additional diagnostic experiments investigated why the CAD-informed estimator failed on approximately 31% of ideal-depth trials. Lowering the minimum number of required end-normal candidates from 20 to 5, and varying the minimum cluster size between 5 and 20 points, did not improve the number of accepted estimates. Instead, some trials that were initially rejected for insufficient candidates progressed to a subsequent rejection because no plausible end-face plane could be fitted.

A comparison against ground-truth end-face visibility provided further insight. The diagnostic mask selected points within 5 mm of the ground-truth end-face plane and at least 5 mm inside its cross-sectional perimeter. Of the 63 failed ideal-depth samples, 54 contained no points within this conservative mask. The remaining nine samples contained only 4–20 visible end-face points. Their estimated longitudinal axes were generally close to ground truth, suggesting that insufficient face evidence, rather than axis estimation alone, limited candidate selection.

A further analysis found that estimated surface normals near the end-face perimeter were frequently contaminated by neighbouring surfaces. In contrast, normals estimated more than 5 mm inside the end face were generally better aligned with the true end-face normal. This suggests that limited interior face visibility and neighbourhood-based normal estimation near geometric edges are interacting causes of failure.

These diagnostics use ground-truth poses and end-face membership only for offline analysis. They suggest that limited visible face evidence, rather than axis estimation alone or simple threshold selection, is the main source of failure. Viewpoint selection and boundary-aware normal estimation are therefore promising directions for improvement.

G. Failure Analysis and Alternative Approaches

Global FPFH/RANSAC registration followed by ICP did not recover a reliable pose, likely because the elongated, repetitive geometry provides few distinctive local features. Local ICP converged within the selected 5 mm and 5-degree tolerances in 15 of 16 controlled initialisations, but these used ground-truth initial poses and do not demonstrate autonomous registration. PCA also produced unreliable cross-sectional axes, while percentile trimming removed extreme points useful for estimating the visible object extent.

Partial visibility was another important source of OBB error. Two randomized poses produced translation errors of approximately 0.53 m because only a short portion of the target was visible near one end, strongly displacing the midpoint of the observed point distribution. The relationship between visible span and error was not consistently monotonic, indicating that the OBB midpoint is not a reliable estimate of the full object centre under arbitrary partial visibility.

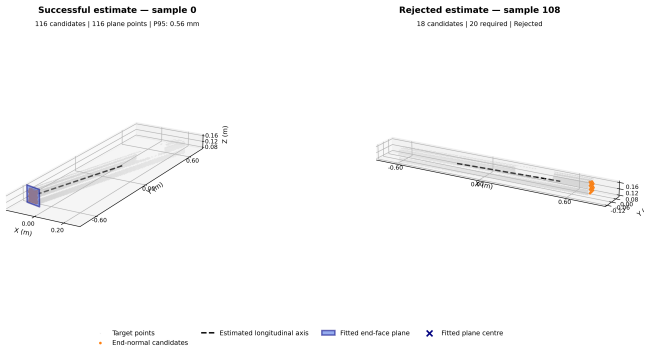


Fig. 5. Examples of the CAD-informed end-face estimator. Left: sample 0, in which 116 end-normal candidates form an accepted end-face plane. Right: sample 108, rejected because its 18 candidates do not meet the minimum threshold of 20. Ground-truth information is used only for offline failure diagnosis and is not provided to the estimator.

VI. Limitations

The sensor model has not been calibrated against a physical D455. Its disparity noise, quantisation, range-dependent dropout, boundary failures, and spatial-correlation scenarios are phenomenological assumptions. Material reflectance, illumination, calibration drift, and proprietary stereo post-processing are not modelled.

The scene uses a cuboid proxy rather than a detailed T-slot extrusion. The randomized dataset varies only horizontal translation and yaw, with a fixed camera. Both estimators use ground-truth segmentation and camera parameters; the CAD-informed method also assumes known target geometry.

Consequently, the experiments isolate selected depth-related effects rather than evaluating an autonomous perception or manipulation system. No physical camera calibration, real RGB-D measurements, or robot grasp executions were performed.

VII. First Six-Month Research Plan

The first six months would prioritise empirical sensor characterisation and manipulation-relevant validation before increasing simulator complexity. The riskiest assumption is that the synthetic depth model captures spatially structured, material-dependent errors relevant to real-world localisation and grasping. Section IV-F shows that OBB estimation errors change measurably with the assumed spatial correlation of depth noise, which has not been measured on a real camera.

TABLE VII
PROPOSED FIRST SIX-MONTH RESEARCH PLAN.

Period	Research activity	Expected outcome
Months 1–2	Establish a reproducible D455 acquisition pipeline. Verify camera intrinsics and RGB–depth alignment, and collect repeated measurements across distances, viewing angles, materials, and illumination conditions. Measure residual disparity autocorrelation and dropout on a flat aluminium target.	A documented real RGB-D dataset and initial measurement-error statistics.
Months 2–3	Compare real observations with synthetic depth. Analyse error distributions, invalid measurement patterns, spatial correlation, and boundary behaviour; fit and validate sensor-model parameters. If measured correlation lengths or dropout rates lie outside the simulated range, revise the model before drawing sim-to-real conclusions.	An empirically informed sensor model with documented uncertainty.
Months 3–4	Expand synthetic scenes across object geometries, poses, camera viewpoints, and illumination conditions. Generate consistent semantic and instance annotations, and evaluate object localisation and 3D reconstruction.	A broader synthetic dataset and baseline perception results.
Months 4–5	Develop a more complete grasp-pose pipeline using CAD geometry, geometric constraints, and collision checking. Evaluate candidate grasps in simulation and, where a suitable robot is available, on the physical system.	Manipulation-relevant robustness results from simulation and, if feasible, physical experiments.
Month 6	Evaluate perception across synthetic and real observations, analyse failure cases, and consolidate the experimental framework.	An initial sim-to-real assessment and prioritised research directions.

VIII. USE OF AI TOOLS

AI assistance was used to support Python development, debugging, interpretation of experimental results, and report preparation. It contributed to discussions about the BlenderProc pipeline, depth-corruption formulation, camera selection, and evaluation methodology. Generated code and suggestions were treated as proposals: implementation choices and reported numerical results were checked against the code and experiment outputs.

One correction concerned the probability model for invalid depth measurements. An initial formulation added the range-dependent and boundary-related failure probabilities directly. Inspection of the implementation showed that the two failure events were sampled independently. The appropriate probability of their union is therefore

$$p_{\text{invalid}} = 1 - (1 - p_{\text{range}})(1 - p_{\text{boundary}}).$$

The implementation and overlap between the failure masks were checked against this formulation.

A second example concerned camera selection. AI assistance initially suggested the Intel RealSense D435 as a practical reference sensor. Rather than accepting a single-camera recommendation, I questioned whether the choice was sufficiently justified and proposed comparing several commercially available RGB-D cameras using a weighted multi-criteria assessment. This led to the evaluation of five candidates and the selection of the D455 based on its highest weighted score for the stated simulation objectives. The final choice was therefore based on an explicit comparison rather than accepting the initial recommendation.

IX. CONCLUSION

This work developed a reproducible pipeline for generating labelled synthetic RGB-D manufacturing scenes and investigating the effect of stereo-inspired depth corruption on geometric pose estimation. The randomized experiments showed that disparity perturbations were the dominant source of degradation for the PCA-based OBB estimator, while spatial correlation produced an estimator-specific trade-off between cross-axis and orientation errors. The CAD-informed estimator achieved more accurate conditional pose estimates but was limited by end-face visibility and estimate coverage. The camera model affected the extremal-point OBB estimator more strongly than the plane-fit CAD estimator: at the 5 mm criterion, OBB success fell from 18.0% to 1.0% under disparity corruption, whereas CAD-estimator success changed only from 60.0% to 58.2% under full corruption. The main limitation is the absence of empirical calibration against a physical RGB-D camera. Future work should therefore prioritise real D455 measurements, validation of the sensor model, and manipulation-relevant experiments.

X. CODE AVAILABILITY

The implementation, experiment scripts, and representative results are available at https://github.com/JoaoZenario/rgb-d_camera_eth. The repository documents dataset generation and experiment reproduction.

REFERENCES

[1] M. Denninger, M. Sundermeyer, D. Winkelbauer, Y. Zidan, D. Olefir, M. Elbadrawy, A. Lodhi, and H. Katam, "BlenderProc: Reducing the reality gap with photorealistic rendering," *arXiv preprint arXiv:1911.01911*, 2019.

[2] K. Khoshelham and S. O. Elberink, "Accuracy and resolution of Kinect depth data for indoor mapping applications," *Sensors*, vol. 12, no. 2, pp. 1437–1454, 2012.

[3] G. Halmetschlager-Funek, M. Suchi, M. Kampel, and M. Vincze, "An empirical evaluation of ten depth cameras: Bias, precision, lateral noise, different lighting conditions and materials, and multiple sensor setups in indoor environments," *IEEE Robotics & Automation Magazine*, vol. 26, no. 1, pp. 67–77, 2019.

[4] Intel, "Intel RealSense Depth Camera D455," https://www.intel.com/content/www/us/en/products/sku/205847/intel-realsense-depth-camera-d455/specifications.html.

[5] Orbbec, "Gemini 2 Stereo Vision Camera," https://www.orbbec.com/products/stereo-vision-camera/gemini-2/.

[6] ——, "Femto Bolt," https://www.orbbec.com/products/tof-camera/femto-bolt/.

[7] Photoneo, "MotionCam-3D M," https://www.photoneo.com/products/motioncam-3d-m/.

[8] Zivid, "Zivid 2+ MR60," https://www.zivid.com/zivid-2-plus-mr60.

[9] Intel RealSense, "librealsense: Intel RealSense SDK 2.0," https://github.com/IntelRealSense/librealsense.